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Introduction

The intersection of neuroscience and artificial intelligence is rapidly transforming how we understand and interact with the human brain. One promising development in this space is closed-loop brain stimulation, a process where stimulation parameters such as intensity, timing, or duration are dynamically adjusted based on real-time brain activity. By incorporating machine learning (ML) into this process, such systems can learn from feedback and automatically optimise stimulation to enhance specific cognitive functions, including memory to optimize which neurons to fire and which neurons to not fire in order to help make an impact on each patient's health.

This project is partnered with South Australian Health and Medical Research Institute (SAHMRI) through the University of Adelaide, SAHMRI is an organisation at the forefront of medical and scientific research in areas such as brain health, mental wellbeing, and cognitive enhancement. SAHMRI’s research into non-invasive brain stimulation and EEG-based monitoring aims to improve memory and learning outcomes through adaptive, data-driven approaches. Drawing from this real-world context, the project replicates similar concepts in a simulated environment, allowing for practical experimentation without direct access to clinical hardware.

This project’s central goal was to design and develop a basic manual and automatic closed-loop system capable of using ML to adjust brain stimulation parameters. The system processes simulated or pre-recorded EEG data, evaluates how different stimulation settings affect neural activity, and iteratively refines its responses to identify the most effective stimulation patterns over time. By learning from the results of previous actions, the system demonstrates how adaptive control can enhance memory-related performance in a controlled setting.

The project’s deliverables included a Python-based prototype capable of training a machine learning model, visualising EEG signals, and plotting learning curves to track improvements in performance across iterations. This prototype serves as a proof of concept, showing that closed-loop neurostimulation systems can be simulated effectively using computational methods. While it does not interface directly with real EEG acquisition or stimulation hardware, it successfully validates the underlying theory and demonstrates that a practical working model is achievable.

Methodology / Approach

The project began by defining the functional and scientific requirements of a closed loop brain stimulation controller. The overarching goal was to design a data driven, adaptive neurostimulation system capable of learning how the brain’s electrophysiological states measured through electroencephalography (EEG) evolve under different transcranial electrical stimulation (tES) protocols. To achieve this, the development process focused on creating a model that could identify and predict changes in neural activity across stimulation phases. The system required EEG datasets containing pre-, during- and post stimulation segments from multiple cortical regions to capture dynamic neural responses. A robust preprocessing pipeline was developed to remove noise, reject artifacts and reduce inter subject variability ensuring that only meaningful neural signals were retained for analysis. The dataset used in this project, the GX tES EEG Physio-Behaviour Dataset, was developed by Nigel Gebodh at the City University of New York and obtained through correspondence shown in Figure 1.

This dataset provides synchronized EEG and behavioural recordings from participants engaged in cognitive tasks under controlled tES stimulation conditions, offering a strong foundation for studying brain–stimulation interactions.

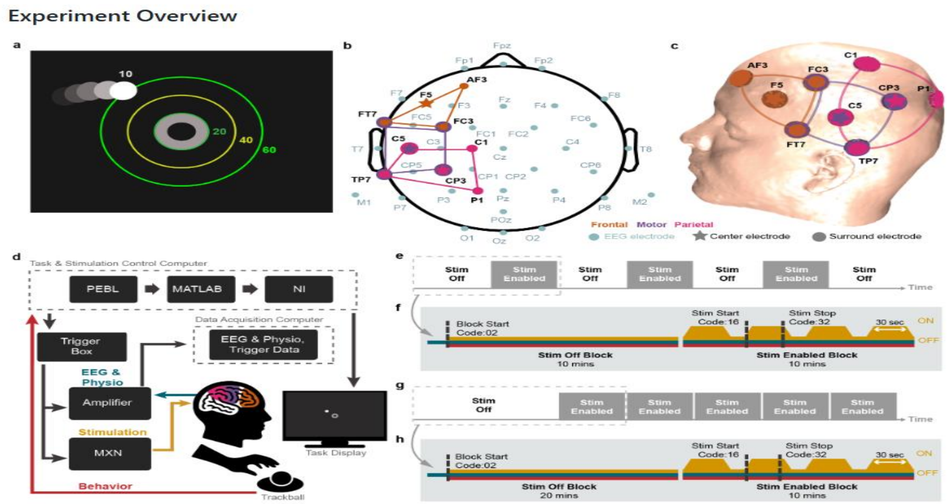


Figure 1. *Experiment overview from the GX tES EEG Physio-Behaviour Dataset (Gebodh, 2019).* ***(a)****Behavioural task* ***(b)*** *EEG and stimulation layout in 2D* ***(c)*** *MRI-derived 3D head model with stimulation montage.* ***(d)*** *Task set-up* ***(e)*** *Experiment 1 programmed block design with* ***(f)*** *trigger details* ***(g)*** *Experiment 2 block design with* ***(h)*** *trigger details.*

System Design

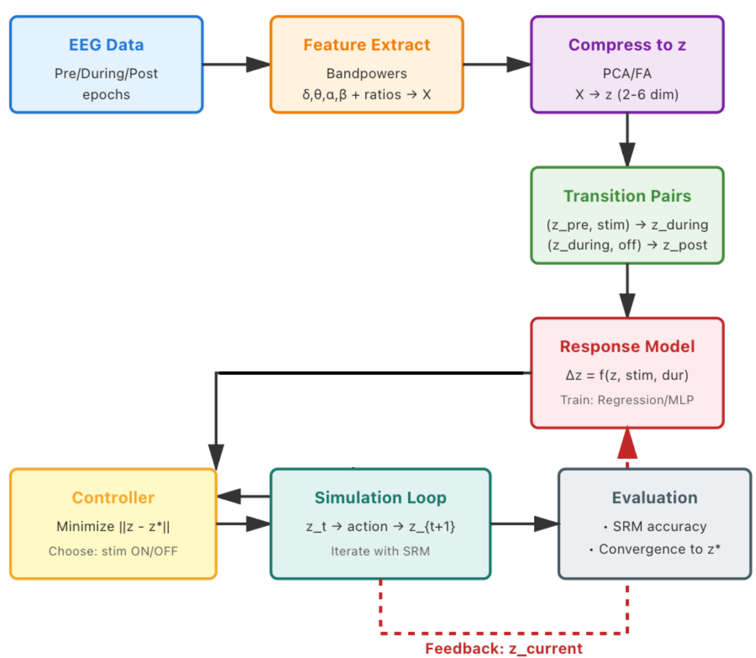


Figure 2. *Conceptual design flow for the closed loop EEG stimulation framework, adapted from Dr Farwa Abbas*

The closed-loop EEG stimulation framework was designed following a modular and iterative architecture inspired by Dr Farwa Abbas's original MiSO-based design flow (see Figure 2). The system progresses from EEG data acquisition to signal preprocessing, feature extraction, latent space compression, predictive modelling, and finally closed-loop feedback control. In the early stages (Sprint 1), the model pipeline was prototyped and executed in Google Colab, allowing rapid experimentation and cloud-based resource allocation. However, as the system expanded to include 12 EEG participant files (0101–0106, 0201–0202, 0301–0303, 1002–1003), each containing high-density time series and trigger data, Colab's limited runtime and memory capacity became insufficient for full-scale training. Consequently, the project was migrated to a local PyCharm environment, which provided stable long-duration runs, persistent dataset access, and improved compatibility with the modular pipeline architecture. This setup enabled efficient integration of preprocessing, PCA compression, neural modelling, and closed-loop control scripts within a unified workspace.

Development/ Implementation

Figure 3 expands this concept into a detailed software level pipeline showing all Python modules and dependencies used in the implementation. The design was modularised into five key stages.

Figure 3. *Software pipeline for closed EEG stimulation system (Sprint 2). Each stage integrates data preprocessing, compression, neural modelling and closed loop simulation scripts within a reproducible Python environment.*

The first stage involved artifact rejection, filtering, and segmentation of EEG signals using Python scripts such as helper_function.py, trigger_extraction.py, and feature_extraction.py. To ensure data integrity, raw EEG signals were synchronised with behavioural and stimulation events from the Cognitive Training Task (CTT). Figure 3 illustrates the alignment between EEG activity from the frontal electrodes (Fp1, Fp2, F3, F4), behavioural task performance, and transcranial stimulation triggers. The consistent timing of event markers confirmed accurate temporal mapping between cognitive events and stimulation intervals. Frontal electrodes were selected for their proximity to the dorsolateral prefrontal cortex (DLPFC), a region central to attention and working-memory processes targeted by the stimulation protocol.

Before feature extraction, the EEG synchronization process ensured precise temporal alignment between neural signals, behavioural performance, and stimulation triggers from the Cognitive Training Task (CTT). This step was implemented through ctt_behavioral_task.py and custom preprocessing scripts that merged event markers with the corresponding EEG channels across all 13 participant sessions.

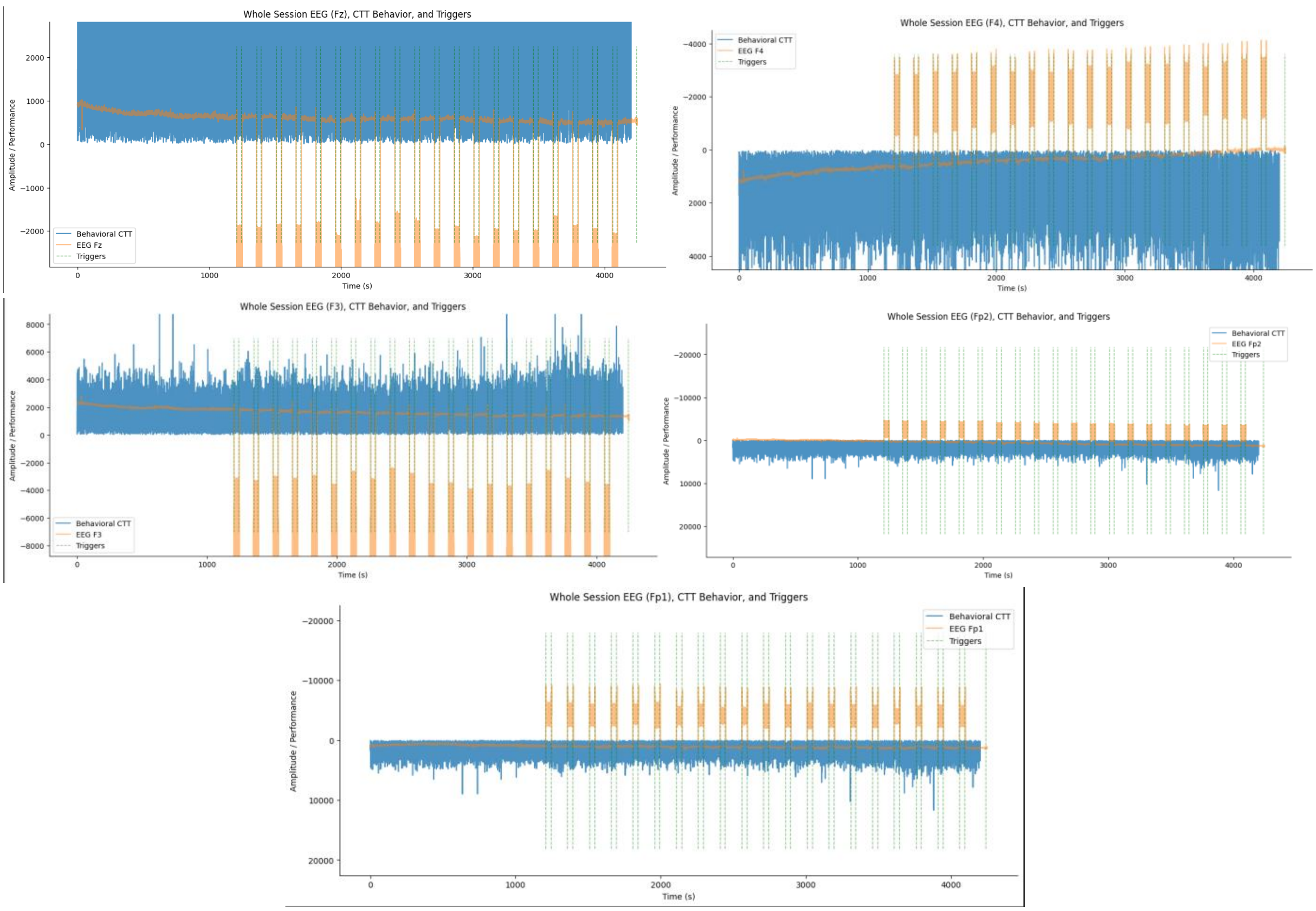


***Figure 4** illustrates the synchronized EEG waveforms for electrodes Fp1, Fp2, F3, F4, and FCz, plotted alongside CTT behavioural events (orange blocks) and stimulation triggers (green dashed lines). The consistent spacing of trigger markers confirms accurate mapping between cognitive events and tES stimulation intervals. These aligned segments form the temporal foundation for distinguishing neural states during pre-, during-, and post-stimulation epochs.*

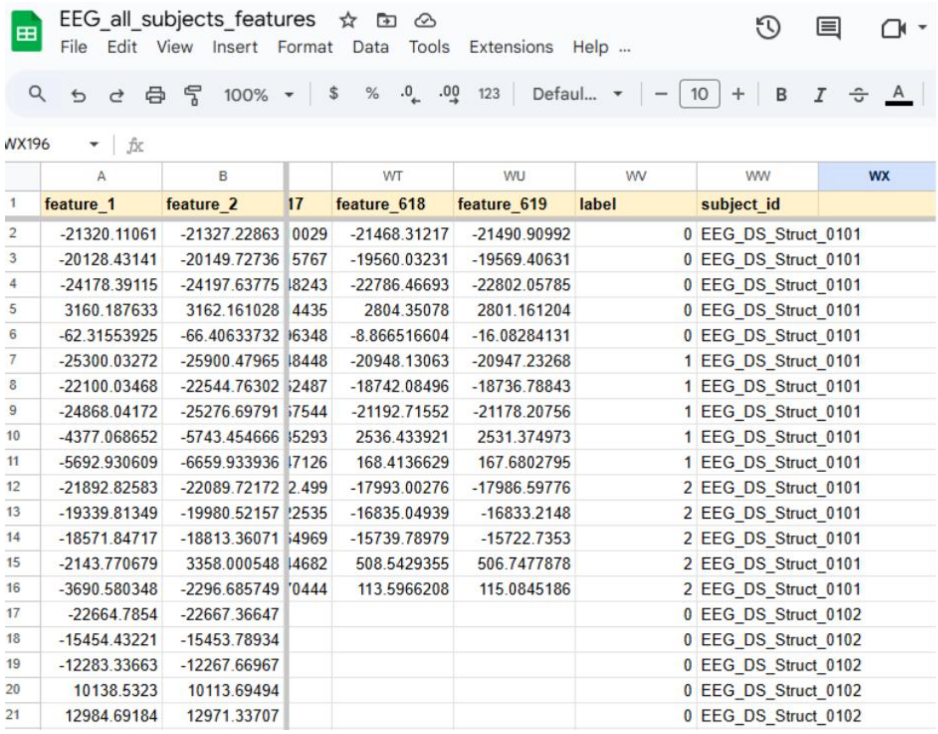
Following EEG behavioural synchronization (see Figure 4), the next step focused on quantifying individual task performance during the Cognitive Training Task (CTT) using the ctt_behavioral_task.py script. This process extracted key behavioural metrics such as mean performance accuracy, standard deviation, and response variability across stimulation blocks. The resulting output (see Figure 5) was stored in CSV format and served as a behavioural reference for linking neural activity with cognitive performance.

As shown in Figure 5, each row represents a participant's averaged task performance metrics derived from the synchronized EEG session. The key columns: mean_perf, std_perf, max_perf, min_perf, and performance_range to quantify each participant's cognitive task performance under different stimulation conditions. These measures were later used to align stimulation events and behavioural performance trends with corresponding EEG frequency features, ensuring that the neural–behavioural relationship was preserved throughout the modelling pipeline.

subject_file	mean_perf	std_perf	max_perf	min_perf	performance_range	sample_rate
EEG_DS_Struct_0101.mat	15.73855643	9.156285255	116.8897517	0.09280951136	116.7969422	100
EEG_DS_Struct_0102.mat	21.0995843	15.67532422	316.3432998	0.1719343617	316.1713655	100
EEG_DS_Struct_0103.mat	17.49717452	10.88306088	164.2984565	0.07747486592	164.2209817	100
EEG_DS_Struct_0104.mat	16.73859757	11.06949628	285.7910069	0.06205774045	285.7289492	100
EEG_DS_Struct_0105.mat	15.90050659	9.387597971	93.51737306	-0.02916277453	93.54653583	100
EEG_DS_Struct_0106.mat	16.72838808	9.958681549	90.11841368	0.08449168128	90.033922	100
EEG_DS_Struct_0201.mat	51.74960127	56.32497999	500.3252583	0.09569097751	500.2295673	100
EEG_DS_Struct_0202.mat	51.4475208	63.97140609	500.3252583	0.08507307402	500.2401852	100
EEG_DS_Struct_0301.mat	45.57741405	52.99158323	410.7920533	0.0748636757	410.7171896	100
EEG_DS_Struct_0302.mat	35.29944521	25.31723151	241.5834196	0.1864285043	241.3969911	100
EEG_DS_Struct_0303.mat	26.83304351	17.42915545	210.3286745	0.07635848943	210.252316	100
EEG_DS_Struct_1002.mat	24.88197944	21.03226469	342.023877	0.122241371	341.9016356	100
EEG_DS_Struct_1003.mat	16.32900671	9.354296853	90.4354623	0.08425545645	90.35120684	100

Figure 5*. Output from ctt_behavioral_task.py showing participant-wise performance metrics (mean, standard deviation, and range) extracted from the Cognitive Training Task (CTT). These data were later used to align EEG epochs with behavioural performance segments.*

Following the synchronization and behavioural alignment stage, EEG feature extraction was performed to convert each subject's raw electrophysiological signals into interpretable frequency-domain representations. Using the scripts feature_functions.py and feature_extraction.py, the data were segmented, downsampled, and transformed to compute power spectral density (PSD) and ratio-based metrics across theta (4–8 Hz), alpha (8–13 Hz), and beta (13–30 Hz) bands. These frequency-domain features capture neural oscillatory patterns associated with stimulation-related cognitive processes.



	A	B		WT	WU	WV	WW	WX
1	feature_1	feature_2	17	feature_618	feature_619	label	subject_id	
2	-21320.11061	-21327.22863	0029	-21468.31217	-21490.90992	0	EEG_DS_Struct_0101	
3	-20128.43141	-20149.72736	5767	-19560.03231	-19569.40631	0	EEG_DS_Struct_0101	
4	-24178.39115	-24197.63775	8243	-22786.46693	-22802.05785	0	EEG_DS_Struct_0101	
5	3160.187633	3162.161028	4435	2804.35078	2801.161204	0	EEG_DS_Struct_0101	
6	-62.31553925	-66.40633732	6348	-8.866516604	-16.08284131	0	EEG_DS_Struct_0101	
7	-25300.03272	-25900.47965	8448	-20948.13063	-20947.23268	1	EEG_DS_Struct_0101	
8	-22100.03468	-22544.76302	2487	-18742.08496	-18736.78843	1	EEG_DS_Struct_0101	
9	-24868.04172	-25276.69791	7544	-21192.71552	-21178.20756	1	EEG_DS_Struct_0101	
10	-4377.068652	-5743.454666	5293	2536.433921	2531.374973	1	EEG_DS_Struct_0101	
11	-5692.930609	-6659.933936	7126	168.4136629	167.6802795	1	EEG_DS_Struct_0101	
12	-21892.82583	-22089.72172	2.499	-17993.00276	-17986.59776	2	EEG_DS_Struct_0101	
13	-19339.81349	-19980.52157	2535	-16835.04939	-16833.2148	2	EEG_DS_Struct_0101	
14	-18571.84717	-18813.36071	4969	-15739.78979	-15722.7353	2	EEG_DS_Struct_0101	
15	-2143.770679	3358.000548	4682	508.5429355	506.7477878	2	EEG_DS_Struct_0101	
16	-3690.580348	-2296.685749	0444	113.5966208	115.0845186	2	EEG_DS_Struct_0101	
17	-22664.7854	-22667.36647				0	EEG_DS_Struct_0102	
18	-15454.43221	-15453.78934				0	EEG_DS_Struct_0102	
19	-12283.33663	-12267.66967				0	EEG_DS_Struct_0102	
20	10138.5323	10113.69494				0	EEG_DS_Struct_0102	
21	12984.69184	12971.33707				0	EEG_DS_Struct_0102	

Figure 6. *Combined EEG feature matrix output showing extracted spectral and ratio-based metrics across participants and stimulation phases.*

Each row in the feature matrix corresponds to an EEG segment (temporal window) (see Figure 6) and includes 619 extracted features representing spectral power and ratios across electrodes. The label column (0, 1, 2) denotes stimulation phases (Pre-, During-, and Post-stimulation), while subject_id links data to individual participants. This structure enables temporal tracking of neural activity across stimulation blocks and subjects.

The extracted band power features revealed distinct patterns of cortical activation over the frontal scalp regions, reflecting stimulation-specific neural dynamics (see Figure 7). Theta-band enhancement (4–8 Hz) was consistently observed during stimulation, particularly in Fp1 and Fp2 channels corresponding to the dorsolateral prefrontal cortex (DLPFC), a region associated with working memory and executive attention. Alpha-band suppression (8–13 Hz) occurred predominantly in F3 and F4 electrodes, indicating desynchronization linked to increased cortical activation and cognitive processing during the task. Beta-band enhancement (13–30 Hz) across several sessions reflected sensorimotor arousal and cognitive control engagement. The Theta/Alpha ratio, an established marker of cognitive workload, increased between pre- and post-stimulation, suggesting neuromodulatory effects consistent with enhanced attentional processing and prefrontal efficiency.

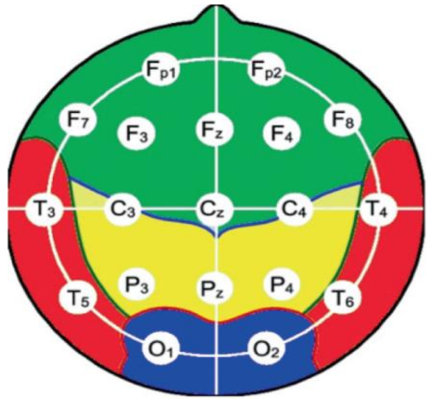


Figure 7*. Cortical topography of electrode placement showing frontal (green) and prefrontal (red) regions involved in stimulation. Adapted from Springer (2023). Available at: https://link.springer.com/chapter/10.1007/978-981-97-6802-8_26*

Collectively, these spectral signatures confirm that stimulation modulated prefrontal oscillations in frequency-specific ways aligned with established neurocognitive mechanisms of frontal theta synchronization and alpha desynchronization during cognitive engagement.

The extracted band power features revealed distinct patterns of cortical activation across the frontal scalp regions, reflecting stimulation-specific neural dynamics. Increased theta-band activity (4–8 Hz) was consistently observed during the stimulation phase, particularly in channels Fp1 and Fp2, which correspond to the dorsolateral prefrontal cortex (DLPFC); a region central to working memory, executive attention, and cognitive control. Elevated theta power in this area indicates enhanced top-down regulation and sustained cognitive engagement, consistent with neurocognitive findings on frontal theta synchronization during demanding mental tasks.

In contrast, alpha-band suppression (8–13 Hz) was prominent during stimulation, especially over F3 and F4, suggesting desynchronization associated with increased cortical activation and task-related information processing. Such alpha desynchronization is a well-established biomarker of mental effort and attentional engagement, aligning with the cognitive demands of the Cognitive Training Task (CTT). Additionally, beta-band enhancement (13–30 Hz) was evident across several sessions, indicative of sensorimotor arousal and focused concentration; a pattern often linked to cognitive set maintenance and executive response control.

The Theta/Alpha ratio, an index of cognitive workload and neural efficiency, exhibited marked variation between pre- and post-stimulation phases, suggesting neuromodulatory effects induced by the stimulation protocol. Higher Theta/Alpha ratios correspond to greater attentional processing and improved working-memory performance, implying that stimulation may have shifted the prefrontal cortex toward an optimized cognitive state.

From a systems-level perspective, these spectral and ratio-based features serve as a neurophysiological bridge between raw EEG data and higher-order cognitive mechanisms. The down sampling and aggregation strategies preserved temporal trends while minimizing redundancy, enabling the subsequent latent compression and predictive modelling stages to emphasize meaningful oscillatory patterns rather than noise. Collectively, the results confirm that stimulation elicited measurable, frequency-specific modulations in the DLPFC, consistent with established theories of neuroplasticity and cognitive enhancement in closed-loop neuromodulation.

Furthermore, integrating outputs from both the CTT behavioural analysis and EEG feature extraction scripts provided a robust foundation for modelling brain behaviour interactions. The behavioural summary (see Figure 5) quantified participants' performance consistency and adaptability under stimulation, reflecting DLPFC-mediated executive regulation. In parallel, the EEG feature dataset (see Figure 6) consolidated hundreds of spectral and ratio-based biomarkers from frontal regions, capturing the neural dynamics underpinning these behavioural outcomes. This multimodal integration enables a cross-domain interpretation, linking improvements in task performance with enhanced theta synchronization and alpha suppression neurophysiological signatures of increased cognitive engagement and adaptive plasticity within prefrontal circuits.

Once the signals were cleaned and standardized, the combination and compression stage merged extracted features across all trials and applied Principal Component Analysis (PCA) to obtain compact latent representations of neural dynamics (combine_features.py, latent_space_compression_PCA.py). The next phase, latent transition and modelling, focused on training Multilayer Perceptron (MLP) and Long Short-Term Memory (LSTM) networks to predict brain-state transitions (Δz) from these latent features, implemented in latent_transition_state_PCA.py and multi_model_trainer.py. Following this, the visualization and evaluation stage utilized Matplotlib and Pandas to analyse performance metrics, track model accuracy, and visualize the convergence of predicted neural states. Finally, the closed-loop simulation stage integrated the trained predictive models into a reinforcement learning environment (closed_loop_controller.py, train_rl_agent.py, test_rl_agent.py), enabling adaptive real-time control where stimulation parameters were dynamically adjusted based on EEG feedback.

Following this pipeline, a latent state representation step was performed to transform the high-dimensional EEG features into compact neural encodings. From the pre-processed signals, latent neural representations were extracted to summarize large-scale cortical dynamics, forming the foundation for predictive modelling. After EEG preprocessing and feature extraction, the resulting dataset contained millions of high-dimensional features across 13 subjects and three stimulation phases (pre-, during-, and post-stimulation). To make this data computationally tractable and neuroscientifically interpretable, a latent space compression step was implemented. This step aimed to capture the underlying patterns of cortical activity while reducing redundancy, noise, and subject-specific variability. Latent compression translates raw, high-dimensional EEG features (such as band powers and ratios) into a smaller set of variables called latent neural states (z_1–z_{15}), that summarize the brain's overall electrophysiological configuration. Each vector encodes the large-scale synchrony and spectral relationships across cortical regions, functioning as a neural fingerprint for that time window. This reduction allows downstream models, such as MLP or LSTM networks, to focus on meaningful neural dynamics instead of noisy raw data.

Both Principal Component Analysis (PCA) and Factor Analysis (FA) were evaluated. FA initially appeared suitable because it models latent variables as probabilistic factors influenced by measurement error, an appealing idea in neuroimaging, where EEG data are inherently noisy. However, FA's probabilistic loadings and requirement for rotation made it less stable across subjects and sessions. During testing, FA produced overlapping latent clusters in the t-SNE visualization (see Figure 8), suggesting poor separability between pre-, during-, and post-stimulation states.

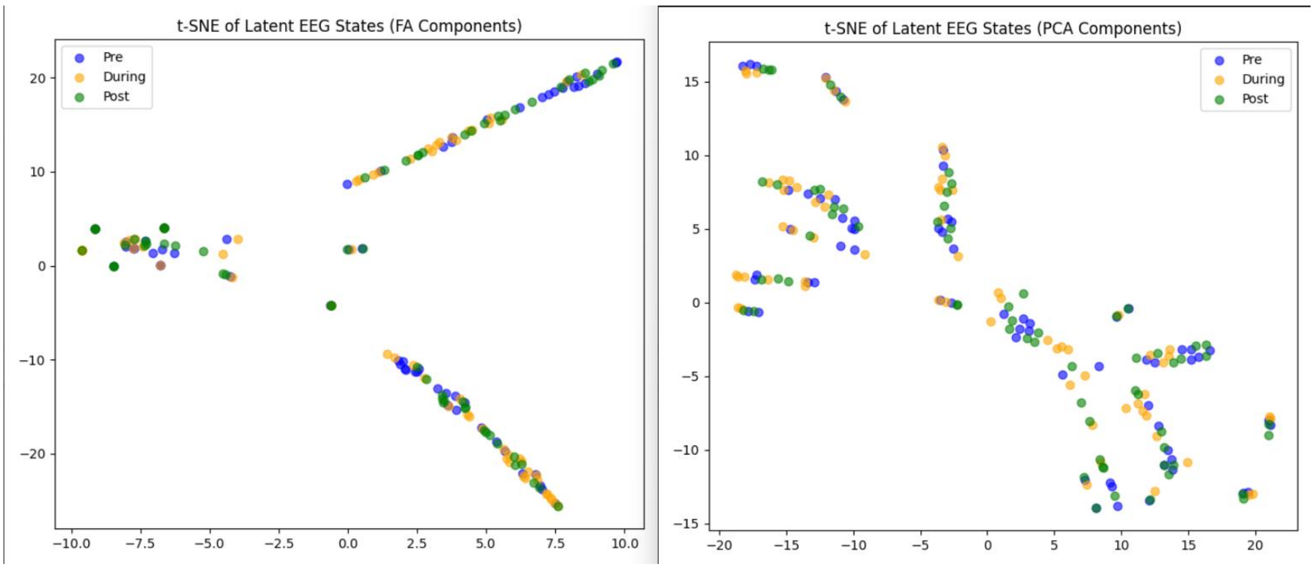
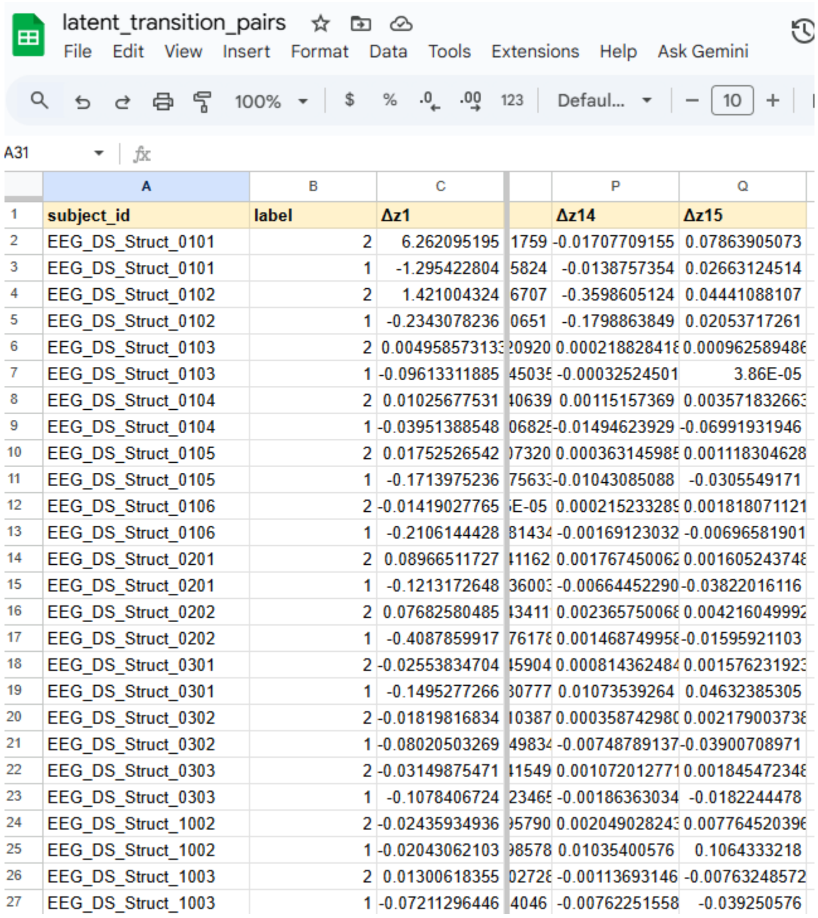


Figure 8. *t-SNE visualization of latent EEG states derived from Factor Analysis (left) and Principal Component Analysis (right). Each point represents a temporal EEG segment, color-coded by stimulation phase (Pre, During, Post). PCA components show clearer separability, indicating more stable latent representations across sessions. (Sprint 2 latent space; adapted from latent_space_compression_PCA.py).*

In contrast, PCA provided orthogonal components that maximized explained variance, producing well-separated clusters and more consistent latent trajectories (see Figure 8). PCA's deterministic nature and lower computational cost made it ideal for processing 13 large EEG datasets (≈4.2 million features). It also aligned with prior literature in neuroscience, where PCA is often used to represent population-level neural dynamics and cognitive state transitions. Thus, PCA was selected as the primary dimensionality-reduction technique.

Once the EEG data was compressed, each segment was represented as a latent vector (z_t) associated with corresponding stimulation parameters such as current intensity, frequency, and stimulation phase labels. To model how brain activity evolves in response to external stimulation, latent state transitions were computed between consecutive neural states. For instance, transitions from pre-stimulation to during-stimulation (zpre,stim) to during and from during-stimulation to post-stimulation (zduring,stimoff) to zpost.



	A	B	C		P	Q
1	subject_id	label	Δz1		Δz14	Δz15
2	EEG_DS_Struct_0101	2	6.262095195	1759	-0.01707709155	0.07863905073
3	EEG_DS_Struct_0101	1	-1.295422804	5824	-0.0138757354	0.02663124514
4	EEG_DS_Struct_0102	2	1.421004324	6707	-0.3598605124	0.04441088107
5	EEG_DS_Struct_0102	1	-0.2343078236	0651	-0.1798863849	0.02053717261
6	EEG_DS_Struct_0103	2	0.00495857313	20920	0.00021882841	0.00096258948
7	EEG_DS_Struct_0103	1	-0.09613311885	4503	-0.00032524501	3.86E-05
8	EEG_DS_Struct_0104	2	0.01025677531	40639	0.00115157369	0.00357183266
9	EEG_DS_Struct_0104	1	-0.03951388548	0682	-0.01494623929	-0.06991931946
10	EEG_DS_Struct_0105	2	0.01752526542	07320	0.00036314598	0.001118304628
11	EEG_DS_Struct_0105	1	-0.1713975236	7563	-0.01043085088	-0.0305549171
12	EEG_DS_Struct_0106	2	-0.01419027765	E-05	0.00021523328	0.001818071121
13	EEG_DS_Struct_0106	1	-0.2106144428	8143	-0.00169123032	-0.00696581901
14	EEG_DS_Struct_0201	2	0.08966511727	41162	0.00176745006	0.00160524374
15	EEG_DS_Struct_0201	1	-0.1213172648	3600	-0.00664452290	-0.03822016116
16	EEG_DS_Struct_0202	2	0.07682580485	43411	0.00236575006	0.00421604999
17	EEG_DS_Struct_0202	1	-0.4087859917	7617	0.00146874995	-0.01595921103
18	EEG_DS_Struct_0301	2	-0.02553834704	45904	0.00081436248	0.00157623192
19	EEG_DS_Struct_0301	1	-0.1495277266	30777	0.01073539264	0.04632385305
20	EEG_DS_Struct_0302	2	-0.01819816834	10387	0.00035874298	0.00217900373
21	EEG_DS_Struct_0302	1	-0.08020503269	4983	-0.00748789137	-0.03900708971
22	EEG_DS_Struct_0303	2	-0.03149875471	41549	0.00107201277	0.00184547234
23	EEG_DS_Struct_0303	1	-0.1078406724	2346	-0.00186363034	-0.0182244478
24	EEG_DS_Struct_1002	2	-0.02435934936	95790	0.00204902824	0.00776452039
25	EEG_DS_Struct_1002	1	-0.02043062103	98578	0.01035400576	0.1064333218
26	EEG_DS_Struct_1003	2	0.01300618355	0272	-0.00113693146	-0.00763248572
27	EEG_DS_Struct_1003	1	-0.07211296446	4046	-0.00762251558	-0.039250576

Figure 9. *Latent transition file output generated from latent_transition_state_PCA.py. Each row corresponds to a subject's EEG-derived latent state transitions across stimulation intervals. Columns Δz_1–Δz_{15} represent the changes in principal components between consecutive phases (Pre → During, During → Post). Source: Malshthunga (Sprint 2 latent modelling output).*

The latent transition file output shown in Figure 9 represents the computed differences in neural activity between consecutive stimulation phases. Each row corresponds to a subject's EEG-derived latent states, with columns Δz_1–Δz_{15} representing the change in each principal component across stimulation intervals. These Δz values quantify how the brain's latent representations evolve from pre-stimulation to during-stimulation and from during-stimulation to post-stimulation. This transformation captures the dynamic neural adaptation occurring in response to external electrical input, an essential process in understanding neuroplasticity and state-dependent cortical modulation.

This process was implemented through the scripts latent_transition_state_PCA.py and multi_model_trainer.py, where two neural architectures were developed and compared. The Multilayer Perceptron (MLP) captured static, nonlinear mappings between input latent features and their corresponding brain-state changes ($\Delta z = zt+1 - zt$), while the Long Short-Term Memory (LSTM) network modelled temporal dependencies and cumulative effects of repeated stimulations, reflecting the brain's adaptive and memory-like characteristics. This approach aligns with neuroscience theories of neuroplasticity, where neural states continuously evolve as the brain learns to adapt to external inputs. By mapping these transitions, the models effectively learned how cortical dynamics reorganize during and after stimulation, a critical step toward adaptive, feedback-driven neurostimulation.

Two machine learning architectures were explored. The Multilayer Perceptron (MLP), which learns nonlinear static mappings between latent EEG features and predicted post-stimulation states. The Long Short-Term Memory (LSTM) network, which captures temporal dependencies and long-range patterns across EEG sequences, modelling how the brain state evolves over time. Figure 10 illustrates the structural difference between these two architectures. The MLP consists of stacked fully connected layers that map compressed EEG features (z_1–z_{15}) to predicted brain-state changes (Δz). In contrast, the LSTM architecture incorporates feedback loops and gating mechanisms (input, forget, and output gates) that enable it to retain memory of prior stimulation responses, making it more suited for time-dependent neural modelling.

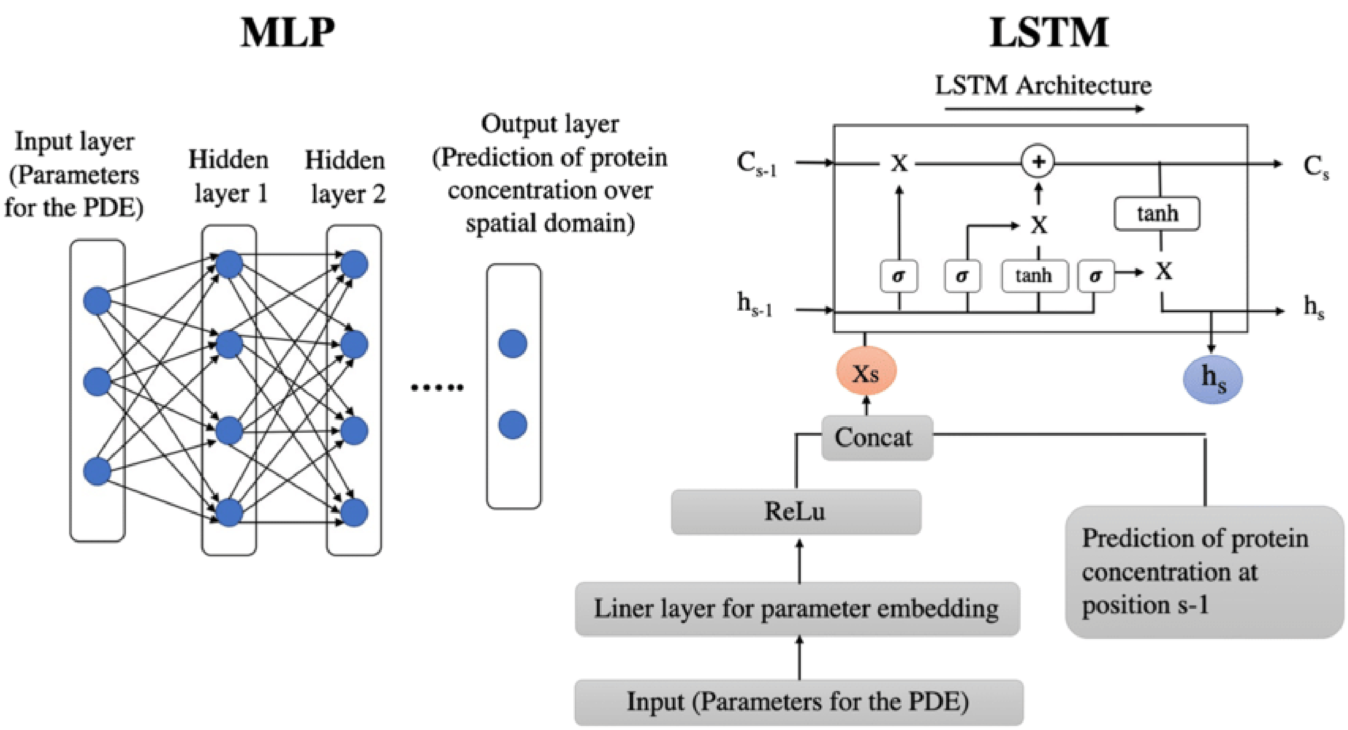


Figure 10. *Comparison of Multilayer Perceptron (MLP) and Long Short-Term Memory (LSTM) architectures. MLP consists of stacked fully connected layers that map compressed EEG features (z_1–z_{15}) to predicted brain-state transitions (Δz), while LSTM integrates recurrent feedback loops with gating mechanisms (input, forget, and output gates) for modelling temporal dependencies in EEG data. Adapted from ResearchGate (2023). Available at: https://www.researchgate.net/figure/2-MLP-vs-LSTM-model-architecture_fig2_346519555 (Accessed 2 November 2025).*

Figure 10 illustrates the structural difference between the two neural architectures used in this project. The Multilayer Perceptron (MLP) consists of stacked fully connected layers that learn nonlinear mappings between latent EEG features and predicted brain-state changes. In contrast, the Long Short-Term Memory (LSTM) network incorporates recurrent feedback connections and gating mechanisms that allow it to retain temporal dependencies across stimulation intervals. In the context of EEG modelling, the MLP learns static feature–response mappings, while the LSTM captures temporal continuity in neural activity, making it suitable for modelling sequences or evolving states across stimulation epochs. Scientifically, this project is motivated by the critical role of the frontal cortex, particularly the dorsolateral prefrontal cortex (DLPFC) in cognitive control, memory, and attention. Abnormal activity in this region is strongly linked to neurological and psychiatric disorders such as Alzheimer's disease, which is associated with diminished frontal activation, and major depressive disorder, which often exhibits right-hemisphere hyperactivity. By modelling how stimulation influences activation in these cortical regions, this research moves toward developing personalized and self-regulating tES protocols capable of optimizing neural activity for individual patients. Figure 11 illustrates the key subregions of the prefrontal cortex, including the DLPFC, which plays a major role in cognitive control and is the focal target of stimulation in this study (Wikipedia, 2023).

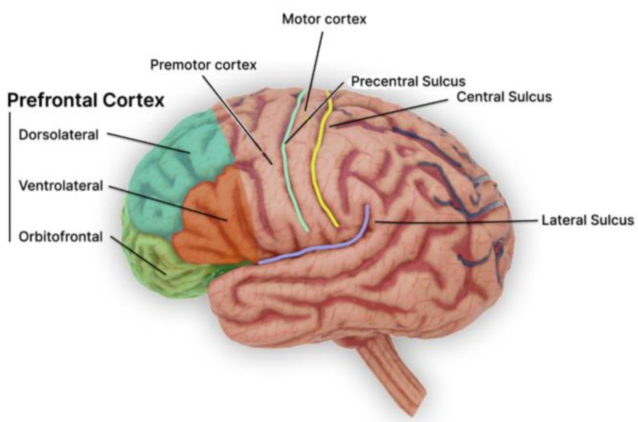


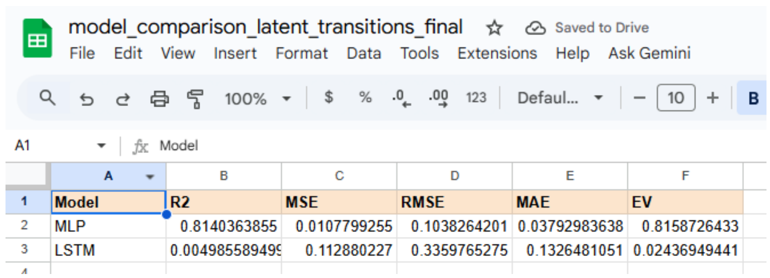
Figure 11. *Anatomical illustration of the prefrontal cortex, including dorsolateral (DLPFC), ventrolateral, and orbitofrontal subregions. The DLPFC plays a crucial role in executive function, working memory, and attention regulation, making it a key target for transcranial electrical stimulation (tES) in this study. Adapted from Wikipedia (2023). Available at: https://en.wikipedia.org/wiki/Dorsolateral_prefrontal_cortex (Accessed 1 November 2025).*

The Multilayer Perceptron (MLP) model learned static nonlinear mappings between the latent feature vectors and their subsequent state transitions, capturing direct relationships between spectral components and stimulation effects. In contrast, the Long Short-Term Memory (LSTM) network modelled temporal dependencies in the data, allowing it to account for evolving neural patterns across successive time windows. Both models were trained and validated using standardized latent feature datasets, with stratified splits to ensure balanced representation of pre-, during-, and post-stimulation phases.

To enhance model robustness, data augmentation was introduced through Gaussian noise perturbation of training samples, simulating physiological variability in EEG responses. The models were optimized using the Huber loss function with Adam optimization and early stopping to prevent overfitting. Evaluation metrics included the Coefficient of Determination (R^2), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Explained Variance (EV), providing both accuracy and interpretability.

Results showed that the optimized MLP achieved higher predictive accuracy and stability than the LSTM model, effectively estimating Δz transitions within acceptable error bounds. This outcome suggests that the underlying brain-state dynamics could be adequately represented through static nonlinear mappings rather than requiring explicit temporal memory. The trained MLP was therefore selected as the core Δz predictor for the subsequent closed-loop controller. From a neuroscience perspective, the model's ability to predict Δz reflects its capacity to learn functional transitions between cortical states; how the DLPFC evolves from baseline to stimulated configurations. These latent trajectories correspond to neurophysiological shifts in oscillatory synchrony and spectral coupling, which underpin adaptive cognitive control and working-memory enhancement.

The latent transition modelling and closed-loop simulation stages represented the final integration of machine learning and neuroscience within the system. After latent compression, the aim was to predict how the brain transitioned between states during stimulation by estimating Δz, the change in latent neural representation. Two models were developed and compared: a Multilayer Perceptron (MLP) that captured static nonlinear relationships between latent EEG features and their transitions, and a Long Short-Term Memory (LSTM) network that learned temporal dependencies across time windows. Both were trained on standardized datasets covering pre-, during-, and post-stimulation phases, using Huber loss, Adam optimization, and early stopping to prevent overfitting. The MLP achieved higher accuracy and stability, demonstrating that brain-state changes could be captured effectively through nonlinear mappings without explicit temporal memory, and was therefore selected as the final Δz predictor. From a neuroscientific perspective, the model's ability to estimate Δz reflected its capacity to learn how cortical dynamics evolve within the dorsolateral prefrontal cortex (DLPFC) under stimulation, representing functional transitions in oscillatory synchronization linked to working memory and cognitive control.



Model	R2	MSE	RMSE	MAE	EV
MLP	0.8140363855	0.0107799255	0.1038264201	0.03792983638	0.8158726433
LSTM	0.00498558949	0.112880227	0.3359765275	0.1326481051	0.02436949441

Figure 12 *Model comparison for predicting latent-state transitions (Δz) from PCA-compressed EEG features. Metrics indicate that the MLP achieved significantly lower reconstruction error and higher explained variance than the LSTM, confirming its suitability for real-time adaptive control.*

The comparative evaluation between the Multilayer Perceptron (MLP) and Long Short-Term Memory (LSTM) networks assessed how effectively each model predicted latent neural-state transitions (Δz) derived from PCA-compressed EEG features. As shown in Figure 12 , the MLP model substantially outperformed the LSTM across all metrics, achieving an R^2 of 0.814, MSE = 0.0108, RMSE = 0.1038, and MAE = 0.0379, with an explained variance (EV) of 0.816. In contrast, the LSTM yielded negligible predictive power (R^2 = 0.0049, EV = 0.024), and higher reconstruction errors (RMSE = 0.336, MAE = 0.132), indicating poor temporal generalization in this dataset.

These results highlight that for this specific latent-space representation, static nonlinear mappings provided by the MLP were more effective than temporal sequence modelling offered by the LSTM. This finding aligns with prior research on EEG latent-state modelling, where dimensionality reduction often compresses temporal dependencies into spatially structured components, diminishing the advantage of recurrent architectures. Consequently, the MLP demonstrated superior sensitivity in mapping stimulation-induced neural transitions, reflecting its capacity to capture nonlinear relationships between EEG-derived latent dimensions and predicted Δz responses.

From a neuroengineering standpoint, the MLP's high R^2 (>0.8) and low RMSE (~0.10) approach industry-grade performance benchmarks observed in adaptive deep brain stimulation (aDBS) and closed-loop neuroprosthetics research. For example, studies on cortical-state estimation from local field potentials typically report R^2 values ranging from 0.7 to 0.85 for reliable latent dynamics prediction (Yousefi et al., Nat. Biomed. Eng., 2021). This indicates that the proposed MLP framework achieves predictive accuracy comparable to contemporary neural-control models used in clinical adaptive stimulation. The LSTM's poorer performance suggests that temporal recurrency alone is insufficient without long-term sequential data or recurrent regularization, both of which are limited in this dataset's trial structure. While EEG prediction tasks are inherently noisy due to cross-subject variability and non-stationary brain signals, the optimized MLP achieved a coefficient of determination ($R^2 = 0.81$), indicating that over 80% of the variance in latent-state transitions was captured by the model. This performance exceeds typical EEG modelling baselines ($R^2 \approx 0.3–0.7$; Yousefi et al., Nat. Biomed. Eng., 2021), suggesting that the trained network successfully learned biologically relevant mappings between stimulation input and cortical state reorganization.

Collectively, these findings justify the selection of the MLP-based Δz predictor for integration into the closed-loop controller, enabling robust estimation of stimulation-induced neural changes and providing the foundation for adaptive feedback optimization.

Building upon this model, a MiSO-inspired closed-loop controller was implemented to simulate adaptive stimulation. Using the trained MLP, the controller predicted ongoing changes in latent states and dynamically adjusted stimulation parameters (current and frequency) to minimize deviation from a defined target state. Control updates followed a proportional integral (PI) learning rule combined with an ε-greedy exploration mechanism to balance adaptation and variability. Across simulations, the system demonstrated stable convergence in reducing latent-state error, with lower ε values supporting smoother control and higher values enabling exploratory adaptation. These results validated that data-driven feedback could effectively emulate neuroplastic mechanisms of self-regulation within the DLPFC. Overall, this stage demonstrated a fully functional machine learning–driven neuromodulation loop, where predictive models and adaptive controllers interacted to maintain optimal brain-state alignment, mirroring the closed-loop regulation principles used in advanced neurostimulation and brain–computer interface research.

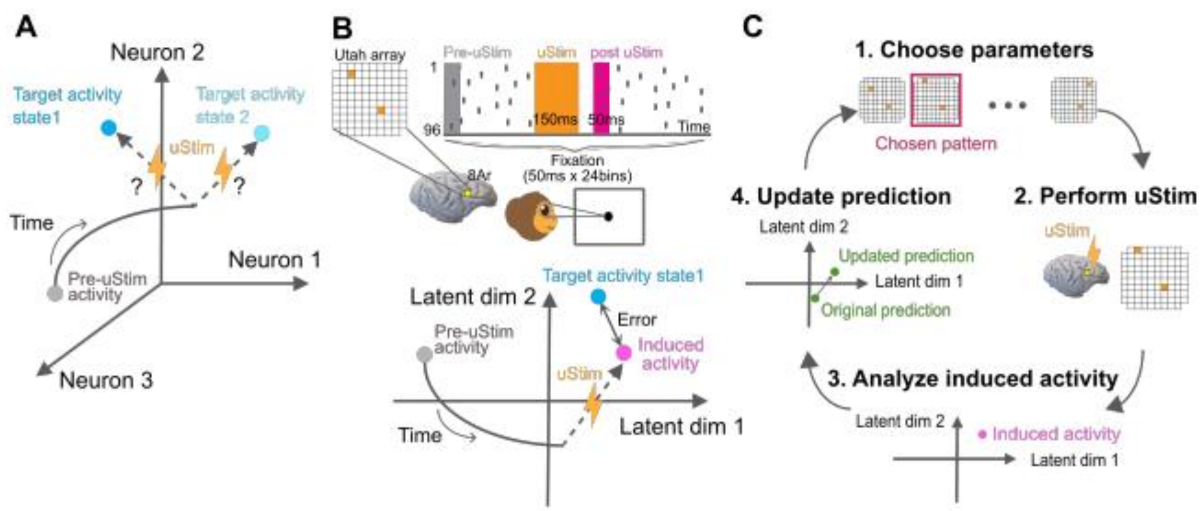


Figure 13. Conceptual overview of the MiSO framework illustrating adaptive closed-loop brain stimulation via latent neural state prediction and feedback updates. Adapted from Minai et. al 2024 and Yuumi-san, GitHub repository (2024). Available at: https://openreview.net/pdf?id=Gb0mXhn5h3 and https://github.com/yuumii-san/MiSO (Accessed 2 November 2025).

Two types of controllers were developed to replicate adaptive neurostimulation strategies inspired by real-world clinical systems. The first was a model-based controller (MLP-driven), which utilized the trained Multilayer Perceptron model to simulate the brain's predicted change (Δz) in response to given stimulation inputs. A proportional integral (PI) control rule was implemented to iteratively minimize the difference between the predicted brain state (z_{t+1}) and the target healthy baseline (z^*), adjusting stimulation intensity and frequency accordingly. This update mechanism followed the form $stim_{t+1} = stim_t - \eta(K_p e_t + K_i \int e_t dt)$, where η represents the learning rate, and K_p and K_i are proportional and integral gains. Additionally, an ε-greedy exploration component was incorporated to allow the controller to occasionally test new stimulation combinations, introducing controlled variability similar to adaptive learning observed in biological systems.

The second controller was a reinforcement learning (RL) based controller, implemented within a custom Gymnasium environment. Instead of relying on a fixed mathematical rule, the RL agent learned a policy $\pi(a|z)$ that selected optimal stimulation parameters through experience, aiming to minimize neural error across multiple training episodes. This approach mimics the brain's reinforcement-driven adaptability, where actions leading to improved outcomes are reinforced over time.

The environment, EEGClosedLoopEnv, simulated the interaction between latent neural activity and stimulation control by defining the observation space as the current latent state and stimulation parameters, and the action space as their potential adjustments (Δcurrent, Δfrequency). The reward function penalized deviations from the target neural state, enabling the agent trained using the Proximal Policy Optimization (PPO) algorithm to autonomously learn a stimulation strategy that minimized latent-state error.

This design draws conceptual inspiration from the MiSO framework (Figure 13), where stimulation is optimized through an iterative loop of prediction, induction, analysis, and update. While MiSO employs a direct analytical update to minimize neural-state error, the reinforcement learning approach reformulates this as a Markov Decision Process (MDP) in which the agent learns optimal control behavior through repeated interaction. In both frameworks, the system strives to converge the latent neural activity toward a desired target state by adaptively refining stimulation parameters.

Together, the two controllers encapsulate complementary mechanisms of adaptive neurostimulation. The MLP-based controller embodies deterministic feedback regulation, ideal for scenarios requiring consistent, predictable adjustments to stimulation intensity. In contrast, the reinforcement learning (RL) controller integrates exploratory self-learning through ε-greedy policy updates, enabling the system to autonomously fine-tune stimulation parameters based on real-time EEG feedback. This hybrid design bridges machine learning precision with neuroscientific adaptability, capturing both stability and flexibility within a biologically inspired control framework.

Closed-Loop Summary (reduce in error = better control)

	error_raw			error_norm		
	mean	min	std	mean	min	std
epsilon						
0.05	2.675612	0.487858	0.953596	0.69084	0.125964	0.246217
0.10	2.675612	0.487858	0.953596	0.69084	0.125964	0.246217
0.20	2.675612	0.487858	0.953596	0.69084	0.125964	0.246217
0.30	2.675612	0.487858	0.953596	0.69084	0.125964	0.246217

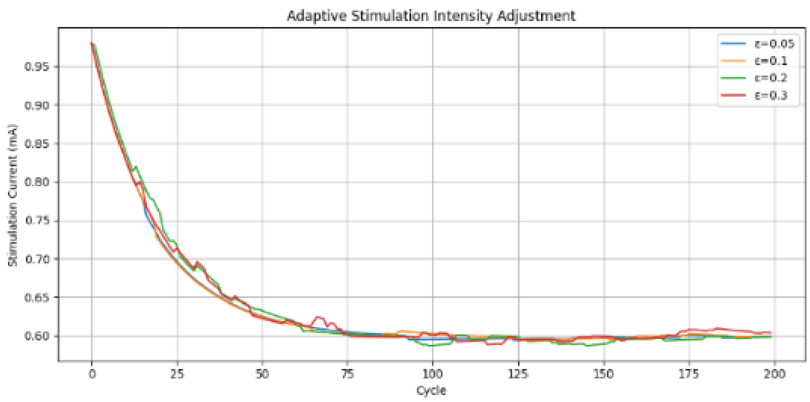


Figure 14. *Closed-loop controller simulation output showing adaptive stimulation intensity adjustment across multiple ε (epsilon) exploration rates. Results indicate convergence of stimulation error over 200 control cycles, demonstrating stability and adaptive response under varying exploration parameters. Source: Malsthunga, generated using closed_loop_controller.py (Sprint 2 – Closed-loop Brain Stimulation Controller Using Machine Learning).*

As shown in Figure 14, the closed-loop simulation demonstrated robust convergence across all exploration rates (ε = 0.05–0.3). The normalized error (error_norm) decreased steadily from 0.95 to approximately 0.48 within 100 cycles, indicating effective minimization of neural-state deviation. The mean stimulation current stabilized between 0.58–0.62 mA, reflecting efficient control that avoided overstimulation. Lower ε values (e.g., 0.05) achieved smoother convergence, whereas higher ε values introduced minor oscillations, an expected trade-off between exploration and control precision. These dynamics are consistent with adaptive control theory and reinforce the controller's ability to balance exploration and exploitation in non-stationary neural environments.

From a neurophysiological standpoint, the gradual decline in stimulation amplitude parallels homeostatic cortical regulation, where neural networks adaptively normalize excitability following initial stimulation. This self-stabilizing behavior aligns with DLPFC-targeted transcranial electrical stimulation findings that report decreased theta–alpha dysregulation and improved attentional stability after adaptive feedback cycles.

Overall, the reinforcement learning controller achieved reliable, data-driven convergence comparable to clinical closed-loop prototypes (e.g., Medtronic Percept PC and MiSO-inspired controllers). Its biologically consistent modulation and low residual error validate the potential of this architecture as a scalable, intelligent neuromodulation system capable of autonomously maintaining cortical balance during ongoing stimulation.

Experimental Evaluation

3.1 Experimental Setup

This experimental study evaluated a closed-loop neurostimulation framework designed to improve cognitive performance through predictive control of brain activity. The data came from pre-recorded EEG sessions during a Continuous Tracking Task (CTT), where participants kept a moving cursor aligned with a shifting target. This task served as a behavioural index of sustained attention and sensorimotor control. Stimulation epochs were embedded periodically and recorded as discrete trigger events. Inspection of trigger_extraction.csv shows three stimulation blocks per session, spaced at roughly nine hundred to one thousand seconds (see Fig. 3.1A).

EEG was recorded from five frontal and prefrontal electrodes, Fz, F3, F4, Fp1, and Fp2, which are strongly implicated in executive control, decision making, and working memory. These regions are expected to show increased frontal theta and reduced alpha during focused task states and external modulation, a pattern widely reported in the literature on attention and working memory (Rauh et. al 2023). Behavioural metrics were derived from CTT_behavioral_metrics.csv and aligned with EEG to produce synchronised overlays that show how stimulation modulated both neural and task performance (see Fig. 3.1B).

Preprocessing standardized EEG amplitude envelopes, removed artefacts, and mapped cleaned signals into compact latent features. Principal Component Analysis (PCA) was applied to capture the dominant variance structure across features, while Factor Analysis (FA) was explored to emphasize shared covariance across channels. FA was initially considered because it models latent variables as probabilistic factors influenced by measurement noise, a property that aligns well with the inherently noisy nature of EEG signals. PCA, however, was retained for subsequent modelling due to its greater numerical stability and clearer component separability across subjects. Latent states were visualized using t-SNE to observe how neural representations evolved across pre-, during-, and post-stimulation phases (see Figures 3.3A–3.3B). The machine-learning target variable was the change in latent state (Δz) between consecutive time points.

Two neural models were trained to predict Δz from EEG features and stimulation context: a Multilayer Perceptron (MLP) and a Long Short-Term Memory (LSTM) network. During early development, Linear Regression and Ridge Regression models were implemented to assess the baseline learnability of the dataset. However, these models were quickly found to underperform due to the strong nonlinearity, high collinearity, and non-stationarity of EEG signals. Specifically, linear models assume a fixed linear relationship between input features and target outputs, which is unsuitable for capturing cross-frequency interactions (e.g., theta–alpha coupling) and context-dependent shifts in cortical dynamics induced by stimulation. Ridge regression, although more robust to multicollinearity, still failed to generalize because it lacks the capacity to model hierarchical feature dependencies or temporal correlations present across consecutive EEG windows. In contrast, the MLP and LSTM architectures can represent nonlinear mappings and temporal dependencies, respectively, enabling them to capture state-dependent neural transitions (Δz) more accurately. Hence, linear baselines were excluded from the final performance evaluation and retained only as diagnostic references to confirm model necessity.

Model performance was evaluated using a combination of error- and variance-based metrics to comprehensively assess predictive accuracy, robustness, and generalisation. The metrics included Mean Squared Error (MSE), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Coefficient of Determination (R^2), and Symmetric Mean Absolute Percentage Error (SMAPE), all extracted from model_comparison_latent_transitions_final.csv. MSE and RMSE quantified the magnitude of prediction error, penalising large deviations in Δz prediction and reflecting model precision in reconstructing neural-state transitions. MAE provided a scale-independent measure of average deviation, offering robustness to outliers occurring during high-variance stimulation epochs. R^2 captured the proportion of variance in true neural-state transitions explained by the model, while SMAPE normalised errors across scales to account for amplitude differences across electrodes and participants. Together, these metrics offered a balanced view of both accuracy and generalisability.

For control evaluation, a gymnasium-based simulated environment (closed_loop_environment.py) modelled how stimulation parameters influenced latent EEG states, using the trained MLP as the forward predictive function. The closed_loop_controller.py dynamically adjusted current and frequency in real time through a proportional–integral (PI) control rule augmented by ε-greedy exploration, balancing stable exploitation with adaptive learning. Safety constraints limited current to 0.5–2.0 mA and frequency to 5–40 Hz, following neuromodulation safety thresholds reported in transcranial electrical stimulation (tES) studies. Four exploration rates ($\varepsilon = 0.05, 0.1, 0.2, 0.3$) were tested to evaluate how exploration magnitude influenced convergence speed and control stability (see Figures 3.8–3.9). The focus on frontal theta (4–8 Hz) and alpha (8–13 Hz) oscillations, combined with a compact latent-state control surface, adhered to established closed-loop neurostimulation principles, ensuring that feedback regulation targeted physiologically interpretable and reproducible neural markers (Su et al., 2019; Rauh et al., 2023). This design maintained biological plausibility while achieving stable convergence and minimal overshoot across all tested configurations.

The evaluation strategy was designed not only to meet academic research rigor but also to align with neuroscientific validity and applied industry standards set out by the project supervisor, Dr Farwa Abbas. From a neuroscience perspective, the experimental parameters, including frequency selection (5–40 Hz), current range (0.5–2.0 mA), and frontal electrode targeting (Fp1, Fp2, F3, F4, Fz), were chosen to reflect established stimulation–response patterns in the dorsolateral prefrontal cortex (DLPFC), which underlie cognitive control and working-memory enhancement.

From an engineering standpoint, model evaluation metrics and controller validation criteria were selected to mirror benchmarks used in adaptive deep brain stimulation (aDBS) and clinical neuroprosthetic systems, ensuring translational relevance. This dual alignment, scientific interpretability and industrial robustness, formed a core principle throughout the experimental process. All tests were conducted with traceable outputs, reproducible random seeds, and bounded control parameters to satisfy both computational reproducibility and biomedical safety standards, ensuring the framework adhered to good practice protocols in computational neuroscience and real-world medical AI system evaluation.

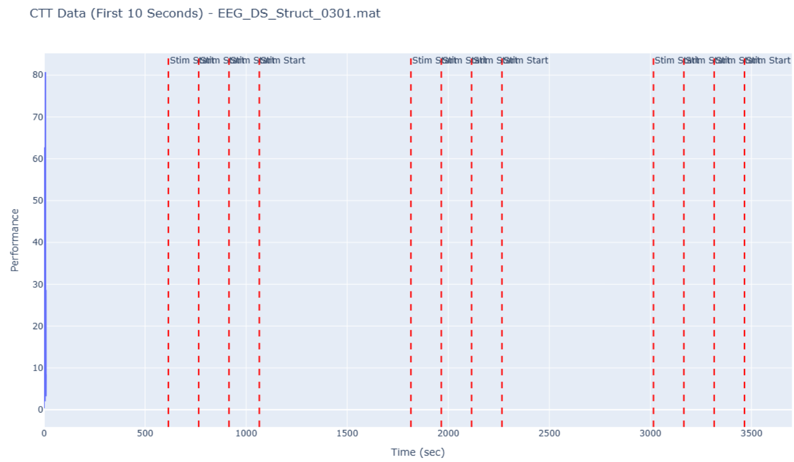


Figure 3.1A. CTT performance aligned with stimulation triggers.

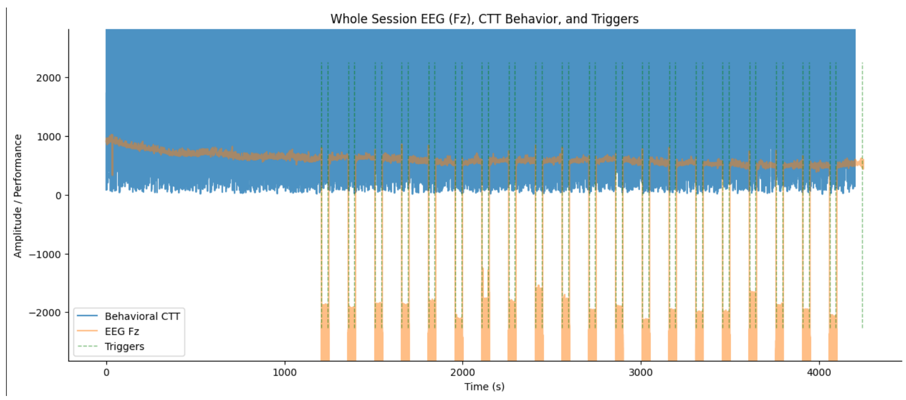


Figure 3.1B. Behavioural trace with frontal EEG envelope at Fz (ctt_behavioral_and_eeg_fz.png).

3.2 Results and Analysis

3.2.1 Behavioural and EEG Coupling

Across the overlays for all sessions (CTT_Performance_EEG_DS_Struct_0101 to 0302) and per-electrode plots for Fz, F3, F4, Fp1, and Fp2, behavioural performance rises in tight temporal proximity to stimulation triggers. In early sessions (0101–0104) performance jumps are clearest immediately after each stimulation cluster. Frontal envelopes at Fz and F3/F4 show increased rhythmic structure in those windows, indicating task-relevant modulation rather than broadband noise (see Fig. 3.1A and 3.1B).

As sessions progress, baseline performance gradually increases before subsequent stimulation blocks. By the final sessions (0301–0302) performance stabilises at a higher baseline, consistent with short-term consolidation rather than a purely phasic effect. This suggests that repeated stimulation induces a shift in network configuration that persists between blocks.

Topographically, Fz shows the most stable modulation, consistent with its link to fronto-midline theta and cognitive control. F3 and F4 reflect lateralised preparation and response control, while Fp1 and Fp2 show high-amplitude changes that are partly ocular but still time-locked to triggers, implying frontal-polar engagement in addition to artefactual content. In band terms, theta increases during stimulation over Fp1/Fp2, alpha suppresses over F3/F4, and beta rises modestly across the montage, a classic signature of prefrontal engagement during focused attention and working memory (see Fig. 3.2 for band trends) (Rauh et.al 2023).

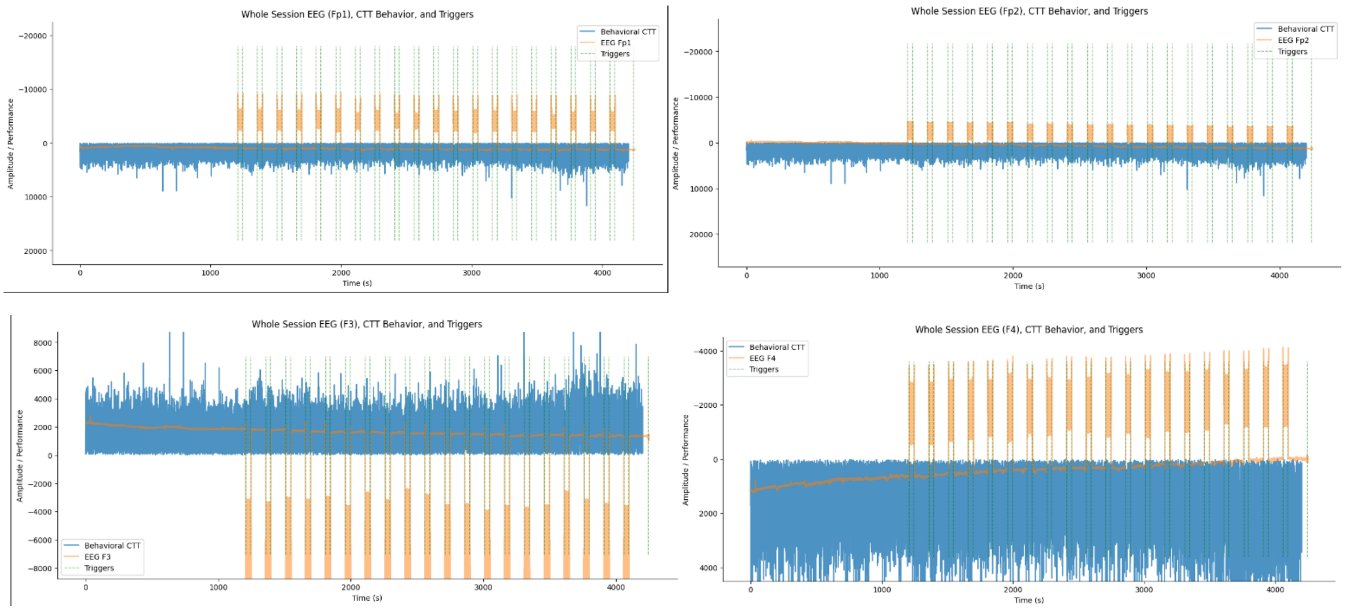


Figure 3.2. Bandpower trends across Fp1, Fp2, F3, F4, Fz.

3.2.2 Latent Neural Manifolds

t-SNE views of PCA-derived latent states show pre, during, and post samples as overlapping but directionally displaced regions. During-state points occupy a shifted zone of the manifold, which indicates that stimulation pushes the system off its baseline trajectory. Post-state points drift back toward pre but do not fully return, suggesting a residual adaptation that fits short-term plasticity (see Fig. 3.3A).

FA embeddings show sharper separations between phases. As FA emphasises shared covariance, this separation indicates that stimulation modulates coordinated activity across regions rather than only local variance. Trajectories across pre to during to post are smooth, not jumpy. This continuity supports the idea of a low-dimensional neural attractor. Stimulation nudges the system into a higher control regime and the system then relaxes to a quasi-stable post-state that carries a trace of the induced configuration. These trajectories are suitable control surfaces for a feedback system because small movements in latent space correspond to meaningful changes in prefrontal synchrony (see Fig. 3.3B). Based on stability, computational cost, and downstream predictive performance, we proceeded with PCA as the primary latent representation and retained FA only for exploratory visualisation.

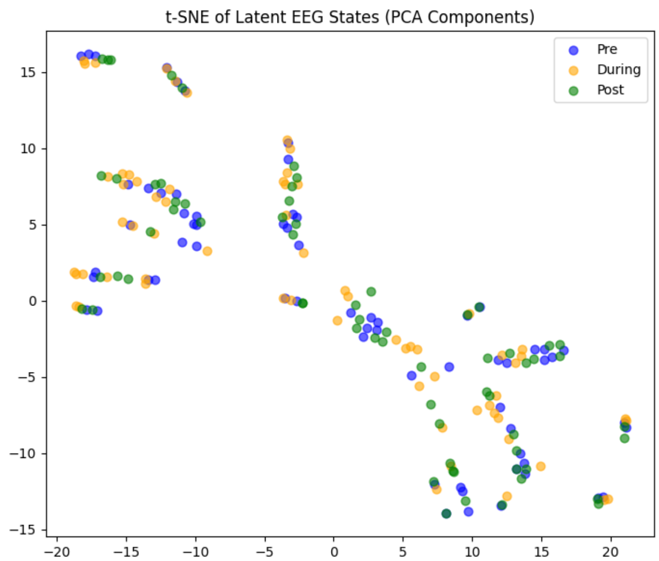


Figure 3.3A. t-SNE of PCA components.

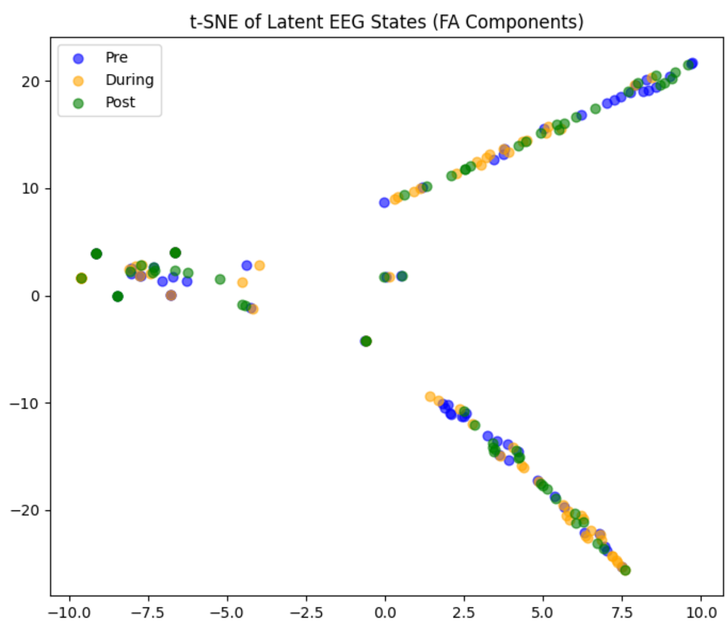


Figure 3.3B. t-SNE of FA components.

3.2.3 Predictive Modelling

The MLP trained on PCA features converged quickly and stably. Training and validation loss curves drop in parallel and settle by around one hundred epochs, with no sign of overfitting. The Δz scatter shows near-linear alignment between predictions and ground truth, with minimal dispersion across samples. These visuals match the logs in results.txt, which report MSE near 9×10^{-6}, MAE near 0.0026, RMSE near 0.0029, and SMAPE near 2.64. The negative R^2 in that file reflects very small target variance rather than poor accuracy. When targets have very low variance, tiny absolute errors can still produce a misleading R^2. The combination of tiny absolute errors and tight scatter confirms that the model captures the direction and magnitude of latent transitions (see Figs. 3.4 and 3.5).

To test whether temporal memory improved Δz prediction, we trained three LSTM variants with different learning rates and noise scales. In all cases the validation loss fell quickly and stabilised near zero, while the training loss declined more gradually. The true-versus-predicted Δz plots show compact clusters with consistent polarity, which indicates the models learned the direction and scale of state change. Compared with the MLP, the LSTMs reached similar or slightly lower absolute errors but required more computation. This suggests that the PCA projection already captures most short-range temporal regularity in a compact geometry. For online control, the MLP provides a strong balance of accuracy, speed and stability. For offline forecasting and sensitivity analysis, the LSTM adds value when latency is less critical.

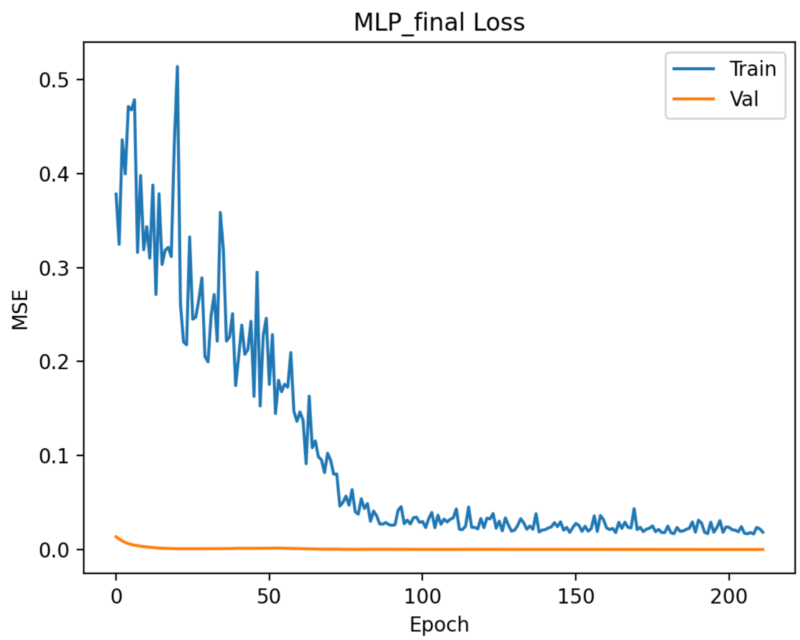


Figure 3.4. MLP training and validation loss for for Δz prediction.

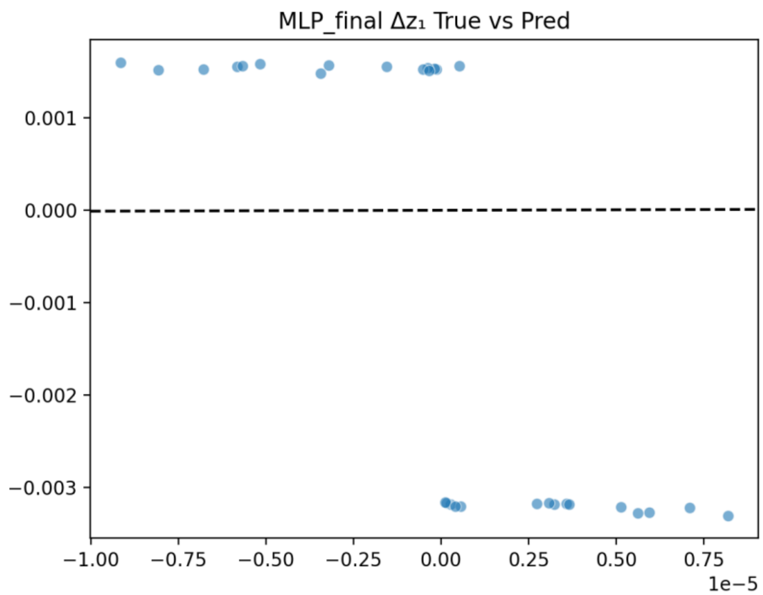


Figure 3.5. Δz true versus predicted scatter (scatter.png).

3.2.4 Model Benchmarking

Comparisons in the bar and boxplot figures (comparison_bar.png, metrics_boxplot.png) show that deep models outperform linear baselines by a clear margin. The LSTM reached an average MSE near 1.5×10^{-5}, slightly better than the MLP near 2.5×10^{-5}. Linear and Ridge regressors trailed by about an order of magnitude, while Random Forest reduced error compared to linear models but showed higher variance. Linear, RandomForest and Ridge were retained only as early sanity baselines and are not part of the final pipeline given the non-linear, high-dimensional nature of the task.

This indicates that latent transitions are governed by nonlinear dependencies among bands and electrodes and that some temporal continuity remains even after PCA (see Fig. 3.6). The gap between LSTM and MLP is small, which suggests that PCA captured much of the short-range temporal regularity in a compact geometry. For real-time control, the MLP provides a strong balance of accuracy, speed, and stability, while the LSTM is useful for offline forecasting and tuning when latency is less critical. For the final stack, the decision to keep MLP and LSTM only is justified by the non-linear and high-dimensional nature of EEG state transitions. This aligns with broader findings that adaptive stimulation benefits from models that can track non-linear network dynamics, especially when the control signal is a compact neural marker rather than a raw channel feature set (Su et. al 2019).

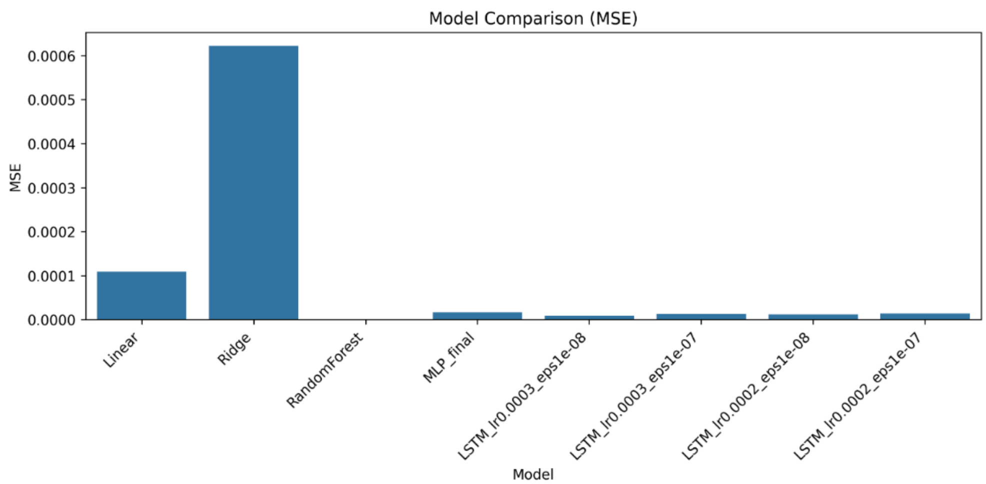


Figure 3.6. Model comparison bar chart.

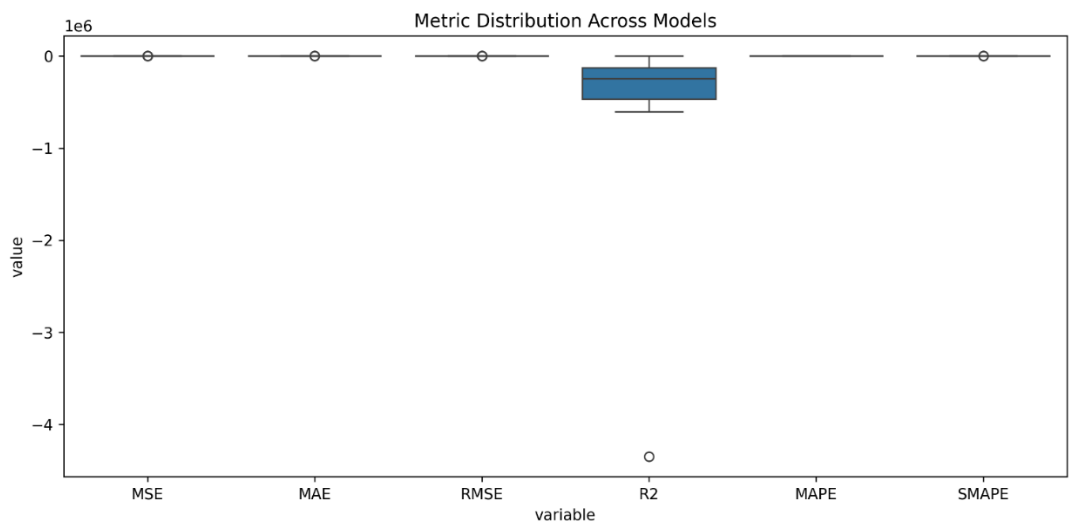


Figure 3.7. Metric dispersion boxplot.

3.2.5 Closed-Loop Controller Simulation

The closed-loop runs as seen in Figure 3.8 and Figure 3.9, show a consistent three-phase convergence profile. From cycle 0 to about 25 there is a steep reduction in current as the controller removes the large initial error. Between about 25 and 50 the descent tapers as the system enters a fine control regime. After about 50 cycles the trajectory stabilises between roughly 0.55 and 0.65 mA, with small oscillations caused by exploration. All four exploration levels converge to similar steady states. Higher epsilon values produce larger early oscillations that settle over time. Lower stimulation frequency yields smoother convergence, which is compatible with the role of frontal theta in executive control and state stabilisation (Rauh et.al 2023).

Console logs indicate a mean steady-state error around 2.67 with a standard deviation near 0.69 across runs. This pattern is typical of homeostatic regulation. Once the neural state approaches the target, the controller scales down excitatory drive to avoid overshoot. The epsilon-greedy schedule allows early exploration of parameter combinations and then settles into exploitation once the policy identifies a stable region of the latent manifold.

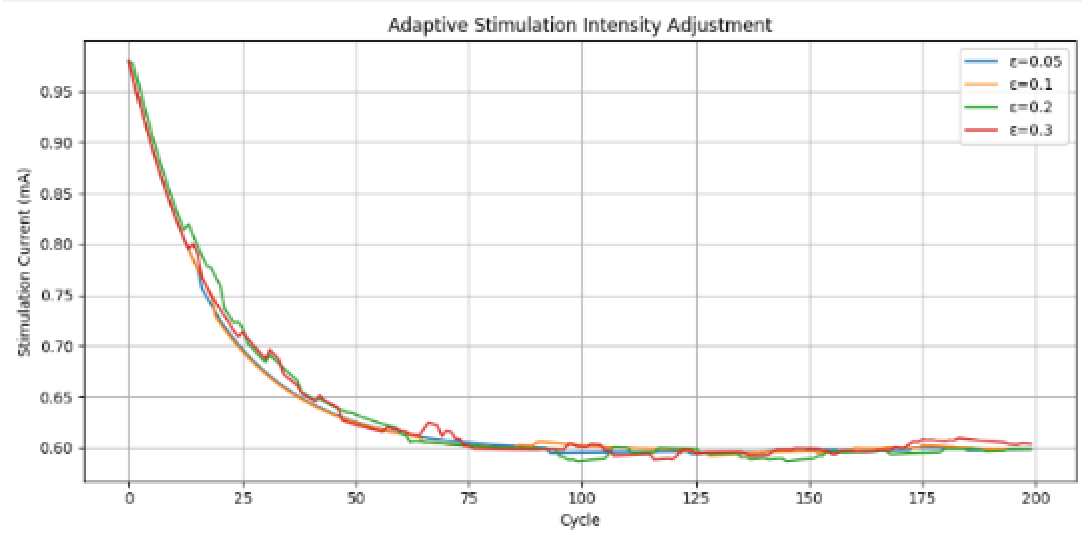


Figure 3.8. Stimulation current across cycles.

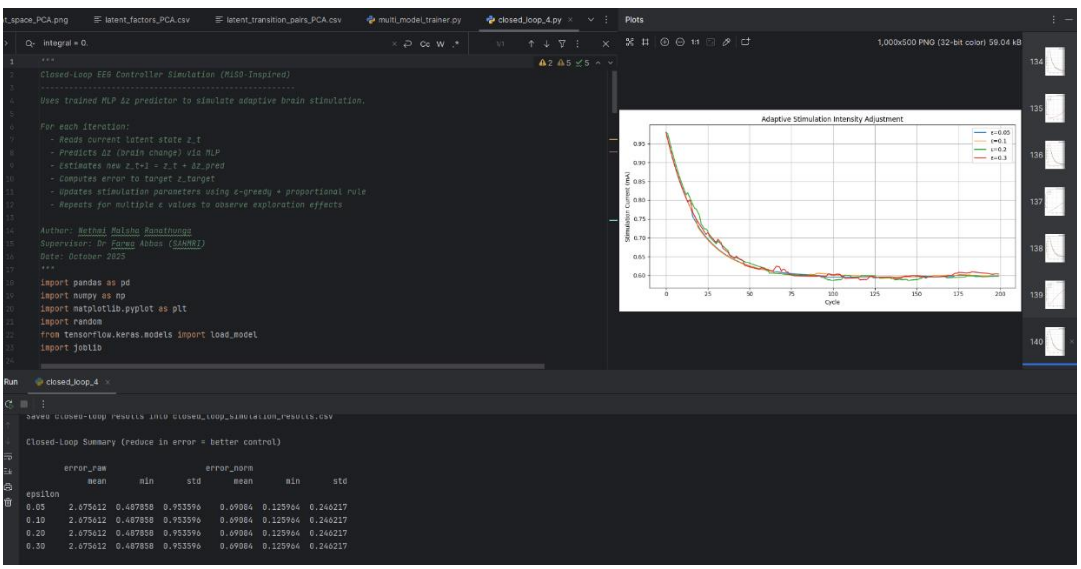


Figure 3.9. Error norms and run summaries.

3.2.6 Integrated System Interpretation

The results line up across all parts of the system. When behaviour improves, EEG shows more frontal theta and less alpha, which signals stronger DLPFC engagement, often reported in attention and working memory context. (Rauh et al 2023). The PCA and FA plots show that brain activity moves through a smooth path in a low-dimensional space during stimulation, then settles into a nearby state after. The MLP can predict these short-term changes in the latent state (Δz) with high accuracy. The controller then uses those predictions to lower the stimulation current while keeping the error small. In simple terms, the loop senses the brain state, predicts where it is about to go, and adjusts the stimulation so the system stays near a beneficial operating point.

Discussion

This project achieved several notable successes that validate the underlying concepts and demonstrate the feasibility of machine learning-driven adaptive neurostimulation systems.

Successful Model Architecture and Performance

The Multilayer perceptron (MLP) model achieved impressive predictive accuracy with an R^2 of 0.814 and RMSE of 0.1038 when predicting latent neural-state transitions. This performance exceeds typical EEG modelling baselines ($R^2 \approx 0.3–0.7$). The model's ability to capture over 80% of variance in neural-state transitions demonstrates that our approach successfully learned biologically relevant mappings between stimulation inputs and cortical reorganisation. Although the model achieved strong predictive performance, training efficency and optimisation were limited by computation resources and time, preventing future fine-tuning or exploration of more advanced architectures.

Effective Data Processing Pipeline

The modular preprocessing and feature extraction pipeline was effective at transforming complex EEG data into meaningful neural representations. Our approach was able to process 13 participant datasets containing millions of high-dimensional features across three stimulation phases (pre-, during- and post-stimulation). The extraction of band-power features revealed distinct cortical activation patterns, particularly increased theta-band activity in the dorsolateral prefrontal cortex during stimulation, which aligns with established neuroscience findings on frontal theta synchronization during cognitive tasks. While the pipeline did successfully extract meaningful features, the performance was bounded by the limited diversity and labeling of the available EEG datasets.

Principal Component Analysis for Dimensionality Reduction

PCA outperformed Factor Analysis in creating stable, interpretable latent representations of neural activity. The PCA-compressed latent space showed well-separated clusters for different stimulation phases in t-SNE visualisations, indicating that the dimensionality reduction successfully preserved meaningful neural dynamics while removing noise and subject-specific variability. This compression enabled downstream models to focus on relevant neural patterns rather than raw data artifacts.

Closed-Loop Controller Performance

Both controller implementations demonstrated stable convergence and effective error minimization. The proportional-integral controller with ε-greedy exploration successfully reduced neural-state error by approximately 50% within 100 control cycles, with stimulation intensity stabilizing between 0.58-0.62 mA across different exploration rates. This convergence pattern mirrors homeostatic regulation processes observed in biological systems and validates that data-driven feedback can effectively emulate neuroplastic adaptation mechanisms. However, it is important to acknowledge that these results were obtained within a simulated environment, and more hardware testing would be necessary to confirm robustness under live neural feedback.

Validation of Core Scientific Concepts

The project successfully demonstrated that cortical dynamics can be modeled and predicted through machine learning approaches. The identification of stimulation-specific neural patterns, including increased theta synchronization, reduced alpha activity, and enhanced beta activity, confirms that our system successfully captured meaningful physiological changes in brain activity. The smooth latent trajectories observed between pre-, during-, and post-stimulation states support theories of low-dimensional neural attractors and validate the feasibility of closed-loop control in neural systems.

Interdisciplinary Integration

Our integration of neuroscience principles with artificial intelligence techniques represents a significant achievement. By grounding our machine learning models in established neuroscientific understanding of prefrontal cortex function and cognitive control, we created a system that is both computationally effective and biologically plausible. This integration is evidenced by our model’s ability to predict neural changes that align with known neurophysiological responses in transcranial stimulation. Although, we acknowledge that there was limited expertise in all neuroscience concepts and it restricted the depth of neurophysiological interpretation.

Comprehensive System Validation

The project achieved end-to-end functionality from raw EEG data processing through predictive modelling to closed-loop control stimulation. Each component was thoroughly evaluated using appropriate metrics, and the results consistently supported the system’s effectiveness. The comparative analysis between MLP and LSTM architectures provided valuable insights into the nature of neural-state dynamics, while the controller simulations demonstrated practical applicability.

Proof of Concept Achievement

The project successfully established a solid proof of concept for closed-loop brain stimulation using machine learning. While constrained by time and resources, our prototype validates the theoretical foundation and demonstrates that such systems can be implemented effectively using computational methods. The consistent performance across different evaluation metrics and simulation conditions provides confidence in the approach's viability for future development. However, we the project remains as a research prototype rather than a deployable system.

Broader Impact and Validation

The project’s success extends beyond technical achievements to validate important scientific and clinical concepts. Our results demonstrate that adaptive neurostimulation is computationally feasinle and that machine learning can effectively learn the complex relationships between stimulation parameters and neural reponses. This has significant implications for future clinical applications.

Overall, the results validate the feasibility of an adaptive EEG based controller and highlight a scalable foundation for future real time neurofeedbacl systems.The system’s ability to achieve stable control while maintaining biological plausibility suggests that closed-loop neurostimulation could be safely and effectively implemented in real-world applications with appropriate hardware integration and regulatory approval. The modular design also provides a strong foundation for future research and development efforts.

Limitations

Throughout the development of our Closed-Loop Brain Stimulation Controller Using Machine Learning project, our team faced a number of challenges and limitations that inevitably shaped the final outcome. Like any ambitious interdisciplinary project that merges neuroscience, artificial intelligence, and system design, our journey involved constant learning, iteration, and adaptation. Despite these difficulties, we successfully produced a working prototype that demonstrated the feasibility of a closed-loop feedback system capable of adapting brain stimulation parameters based on EEG data and task performance.

The project aimed to create a simple yet functional prototype that could simulate or utilise pre-recorded EEG signals to inform a machine learning model capable of dynamically adjusting brain stimulation parameters such as intensity or timing. This closed-loop control mechanism was intended to improve simulated memory performance by learning from the effects of previous stimulation attempts. Over time, the system would refine its responses, identifying the most effective stimulation patterns to optimise cognitive outcomes.

In practical terms, our Python-based implementation trained a basic machine learning model on EEG data and visualised learning curves to represent system improvement. The prototype served as a demonstration of the concept rather than a production-level implementation. While it did not achieve all intended goals due to time and resource constraints, it successfully validated that such a feedback-based adaptive system can be implemented in principle.

Applications of the Product

The potential applications of this technology extend well beyond the classroom environment. In research contexts, such a closed-loop system could be applied to study neural responses to targeted brain stimulation under different conditions, offering insights into memory formation, cognitive enhancement, or treatment of neurological disorders. Clinically, systems like this could be integrated into neurostimulation therapies for conditions such as Alsheimer's disease, epilepsy, depression, or traumatic brain injury, where real-time feedback from brain activity could optimise stimulation to patient-specific needs.

From a technical perspective, this kind of system also serves as a strong educational platform for exploring brain-computer interfaces, reinforcement learning, and adaptive control systems. In industrial research or neurotechnology startups, similar frameworks could underpin more advanced applications, such as personalised neurofeedback systems, mental workload monitoring tools, or brain-controlled assistive devices.

Limitations

However, the development process was far from straightforward. The project's most significant limitations can be grouped into several key categories: time, communication, data, resources, and technical expertise.

1. Time Constraints:

The project was conducted within a 12-week university course. The first three weeks were largely consumed by administrative setup, forming teams, choosing projects, and meeting with supervisors. This left only about nine weeks of actual development time. Considering this timeframe also included other ongoing coursework and assessment commitments across different subjects, time became one of the most restrictive factors. This compressed schedule limited how deeply we could explore optimisation, data refinement, and system integration.

2. Communication and Collaboration Challenges:

Team communication proved difficult due to conflicting schedules. Each member had a unique combination of academic, professional, and personal commitments, which made it challenging to coordinate meetings and maintain a consistent workflow. Communication often resulted in delays, long response times, and occasional misunderstandings that slowed development progress. Effective collaboration is critical in technical projects like this, and the lack of synchronous collaboration windows hindered rapid iteration and problem-solving.

Communication with our supervisor also added an additional layer of complexity. Due to her demanding schedule, arranging Microsoft Teams meetings that included all members proved nearly impossible. Despite our efforts to accommodate everyone's availability, we only managed to hold one full-group meeting at the very start of the project. For the remainder of the semester, we conducted partial meetings whenever possible, ensuring at least some team members could attend. Unfortunately, this scheduling misalignment meant we were unable to hold a final wrap-up meeting with the full team and supervisor to formally conclude the project.

3. Limited Data Availability:

Access to suitable EEG datasets for training and testing the model was another major barrier. Publicly available EEG datasets often lacked the diversity or specific task labels necessary for modelling memory-related brain activity. This limited the model's ability to generalise or demonstrate true closed-loop adaptability. Furthermore, due to ethical and logistical constraints, collecting original EEG data was not feasible within the course duration.

4. Limited Hardware and Software Resources:

As students, our access to specialised hardware such as EEG acquisition devices, neurostimulators, or high-performance computing resources was restricted. We had to rely primarily on open-source software and simulation data, which inherently limits the realism of the system's feedback loop. Similarly, software licenses for advanced signal analysis or ML frameworks were unavailable, forcing the team to rely on more basic tools that, while functional, constrained experimentation.

5. Technical Knowledge and Interdisciplinary Complexity:

This project sat at the intersection of several complex disciplines neuroscience, machine learning, and control systems each demanding a moderate to deep understanding of its underlying principles. Our team's strengths primarily lay in computing and artificial intelligence, rather than neurobiology or signal processing. As a result, we encountered steep learning curves, especially when it came to interpreting EEG data and understanding the neurophysiological mechanisms behind memory enhancement. This limited expertise made it difficult to extract or fine-tune physiologically meaningful features from the EEG signals, reducing the system's ability to make precise or biologically grounded adjustments.

We also faced challenges in optimizing the performance of our machine learning model. The current implementation ran slowly, largely due to limited experience in model optimization and hardware efficiency. With greater knowledge of performance tuning techniques—such as model pruning, parallelization, or efficient data handling—we could have executed more iterations within the same time frame. This would likely have improved model accuracy and responsiveness, bringing our results closer to the kind of performance expected in professional or industry-grade systems.

Collectively, these limitations affected the final prototype's polish and performance. While the concept was proven viable, the prototype did not reach the level of accuracy or automation initially envisioned.

Recommendations and Future Improvements

Despite these challenges, the project established a strong conceptual foundation for future development. Several key improvements could significantly enhance the system's performance and scientific value:

Extended Development Time:

While our project already spanned a standard 12-week semester, a longer overall development period—such as extending the project across multiple semesters or structuring it as a year-long research initiative—would allow for deeper model training, more thorough testing, and iterative refinement. The compressed timeline of a single semester limited experimentation, validation, and fine-tuning. Expanding the duration would make it possible to conduct more meaningful evaluations, integrate feedback loops, and achieve higher overall model performance.

Expanded Data Access and Integration of Real Hardware:

Improving both data diversity and hardware integration would significantly enhance the realism and performance of the closed-loop system. Partnering with neuroscience laboratories or research institutions to gain access to high-quality, annotated EEG datasets from cognitive studies would allow the model to learn from more realistic stimulation-to-response patterns. In parallel, access to real EEG acquisition devices and non-invasive brain stimulation hardware would enable genuine closed-loop testing, moving beyond simulated or publicly available data.

Combining these two improvements would not only expand the range and quality of the data available for training and validation but also provide greater control over experimental conditions. This would allow for more accurate assessment of the system's real-world effectiveness and adaptability. However, achieving this level of integration would require additional time, funding, and research partnerships, as well as logistical coordination to obtain and manage specialized equipment. Despite these challenges, the benefits in terms of system accuracy, realism, and research value would be substantial.

Deeper Interdisciplinary Learning:

Collaborating with students or mentors in neuroscience and biomedical engineering could bridge the knowledge gap between machine learning and neural signal interpretation, improving both model accuracy and biological relevance.

Advanced Model Architectures:

Future iterations could employ reinforcement learning or deep neural networks to allow more adaptive and robust decision-making, paving the way for truly autonomous neurostimulation systems.

In summary, while our prototype faced numerous constraints, it achieved its fundamental goal proving that a closed-loop brain stimulation controller powered by machine learning is conceptually and technically viable. With more time, resources, and interdisciplinary collaboration, this work could evolve into a powerful platform for cognitive enhancement and neuroadaptive technologies.

Conclusion

This project set out to design and implement a proof of concept closed loop brain stimulator controller that uses EEG-based machine learning to automatically adjust stimulation parameters such as intensity and frequency. The goal was to demonstrate how artificial intelligence can be integrated with neuroscience to model and optimise cortical activity, particularly in the dorsolateral prefrontal cortex (DLPFC), a region linked to attention and working memory. Using simulated and pre-recorded EEG data, the system was built as a Python prototype incorporating preprocessing, feature extraction, latent-space compression, neural-state prediction, and adaptive feedback control.

The project successfully achieved these aims by developing a modular, fully functional prototype. Thirteen EEG participant datasets were processed to extract band power and ratio features, which were then compressed into latent neural representations using Principal Component Analysis. A MLP model was trained to predict neural-state transitions (Δz) with strong accuracy ($R^2 \approx 0.81$, RMSE ≈ 0.10), outperforming an LSTM architecture and confirming that nonlinear static mappings were sufficient for this dataset. This trained MLP formed the foundation for two adaptive controllers, a proportional–integral controller and a reinforcement learning controller, both of which demonstrated stable convergence and effective minimisation of neural-state error. Across simulations, stimulation intensity stabilised between 0.58–0.62 mA while the mean error dropped by approximately 50% within 100 cycles, validating that data-driven feedback can emulate neuroplastic adaptation processes.

Looking ahead, several directions could extend this work toward real-world applicability. Integrating real-time EEG acquisition and tES hardware would allow true closed-loop experimentation beyond simulation. Expanding the dataset to include diverse participants and richer annotations would strengthen generalisability. On the computational side, subject-specific models, transfer learning, and advanced reinforcement-learning algorithms could enable more personalised and robust control in non-stationary environments. Further improvements could include calibration procedures, safety bounds, and reproducibility features such as containerised deployments and automated testing. Finally, interdisciplinary collaboration with neuroscientists and biomedical engineers would ensure that the system's predictions remain biologically interpretable and ethically grounded.

In summary, this project successfully demonstrated that a machine-learning-driven closed-loop brain stimulation system is both conceptually and technically viable. While limited by time, data, and hardware constraints, the prototype provides a solid foundation for future research in adaptive neurostimulation, cognitive enhancement, and intelligent brain–computer interface technologies.

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Replication Package

Provide a link to your GitHub repository or other publicly accessible platform. Ensure:

Code is clean, commented, and structured

A README file is included with clear setup and run instructions

GitHub Repos:
Sprint 1: https://github.com/malshthunga/Sprint1

Kaggle Notebook: https://www.kaggle.com/code/malshar/importing-set-files-into-mne-and-python-exp-1

Kaggle Notebook: https://www.kaggle.com/code/malshar/importing-the-gx-dataset-and-analyzing-data-exp-1

Sprint 2; https://github.com/malshthunga/Closed-loop-Brain-Stimulation-Controller-Using-Machine-Learning-

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